

Unifying Special Relativity and Quantum Mechanics

Lecture 26 — The Goldstone Theorem Proved, and Two Notable Breaking Patterns

(Completion of the proof, #Goldstones = $\dim G - \dim H$,
the examples $SO(N) \rightarrow SO(N-1)$ and $SU(N) \rightarrow SU(N-1)$,
and the beginning of the geometric picture of the vacuum manifold)

Typed version of the handwritten notes, with added explanations in blue boxes

Added explanation (not in the original notes) — How to read this document

Conventions. Exactly as in Lectures 1–25: normal text and the yellow / pink / grey boxes are a faithful transcription of the handwritten sheets, and the text colours follow the pens of the original (blue, magenta, green, red, violet, pencil). Everything inside a blue box like this one is material I have added; it is not in the original notes.

Pagination. Still the same chapter, still twelve stars, still no restart. Lecture 24 ran [p. 1*****]–[p. 22*****], Lecture 25 ran [p. 23*****]–[p. 42*****], and this scan picks up at [p. 43*****] and runs to [p. 52*****]. Ten sheets, circled numbers consecutive, no duplicates, nothing missing, nothing out of order. PDF page n of this file is sheet $42 + n$. This is a short lecture.

It begins mid-equation and ends mid-argument. Sheet [p. 43*****] opens with the very equation that Lecture 25 stopped on, evaluated at $\phi = \phi_0$ — so the first two sheets are the conclusion of the proof begun there. At the other end, sheet [p. 52*****] introduces the vacuum manifold $\mathcal{M}_0$ and stops after two green annotations, with the geometric argument plainly unfinished. Section 12 records the fragment as it stands, and a blue box carries it to its conclusion ($\mathcal{M}_0 \cong G/H$, and $\dim \mathcal{M}_0 = \dim G - \dim H$) so you can see the destination.

Two slips flagged in place.

1. On [p. 49*****] the line “ $\Phi^\dagger \Phi = (\rho_1^*, \dots)(\dots)$ ” omits the factor $\frac{1}{2}$ coming from the explicit $1/\sqrt{2}$ in the definition of Φ . What is actually being computed is $2\Phi^\dagger \Phi$, as the boxed result on [p. 50*****] correctly states. The notes self-correct one sheet later; I have written the factor in from the start (Section 9).
2. On [p. 45*****] and [p. 48*****] the potential is written once with $-\frac{\mu^2}{2}\phi^T \phi$ (real case) and once with $-\mu^2 \Phi^\dagger \Phi$ (complex case). These are consistent, not contradictory — the complex field packages two real ones — but the differing factors of 2 propagate into the two vacuum conditions (v^2 versus $v^2/2$), and it is easy to think one of them is wrong. A blue box on [p. 48*****] sets the bookkeeping out explicitly.

Notation carried over. T^a are unbroken generators ($T^a \phi_0 = 0$, spanning $\mathfrak{h}$), X^α are broken ones ($X^\alpha \phi_0 \neq 0$), and T^A runs over both, $A = 1, \dots, \dim G$. The mass matrix is $M_{ac}^2 = \partial^2 V / \partial \phi^a \partial \phi^c(\phi_0)$.

Added explanation (not in the original notes) — Where we left off — recap of Lecture 25

Lecture 25 did three things. It named the two particles of the broken phase — the massive *Higgs* h and the massless *Goldstone* π — and showed that the two descriptions $\mathcal{L}_\theta$ (Cartesian) and $\mathcal{L}_\pi$ (exponential) are related by a *non-linear field redefinition*, which changes the Lagrangian but not the S -matrix. It then drew the crucial distinction: non-renormalizable operators generated this way are *spurious*, and the diagnostic is not the operator dimension but whether the amplitude grows with energy. Computing $\pi\pi \rightarrow \pi\pi$ confirmed the growth cancels, via $s + t + u = 0$.

It then built the effective theory of the Goldstone alone, $\frac{1}{2}(\partial\pi)^2 + c[(\partial\pi)^2]^2$, and matched it onto the full theory to get $c = \lambda/m_h^4$ with a cutoff at $E \sim m_h/\lambda^{1/4}$.

Finally it set up the general problem: a scalar multiplet with a compact global G , the mass matrix M_{ac}^2 , the split of the generators into unbroken (closing a subalgebra $\mathfrak{h}$, giving $G \rightarrow H$) and broken. It differentiated the invariance condition of the potential, obtained $\dim G$ equations, and stopped at the words “we evaluate this equation on the vacuum ϕ_0 ”.

Sheet [p. 43***] of this lecture is that evaluation.** The first two sheets therefore complete the proof; the remaining eight apply it to two families of examples and begin the geometric reformulation.

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PART I — The Goldstone theorem [p. 43*****]–[p. 44*****]

1 Evaluating on the vacuum: the two cases [p. 43*****]

$$\frac{\partial^2 V}{\partial \phi^a \partial \phi^c}(\phi_0) (T^A)^a{}_b \phi_0^b + \underbrace{\frac{\partial V}{\partial \phi^a}(\phi_0) (T^A)^a{}_c}_{=0} = 0$$

We get

$$\frac{\partial^2 V}{\partial \phi^a \partial \phi^c}(\phi_0) (T^A \phi_0)^a = 0$$

for $A = 1, \dots, \dim G$ ★

↪ This is precisely the mass matrix we defined before.

There are two cases:

1) If “ A ” corresponds to an unbroken direction

$$T^A \equiv T^a, \quad T^a \phi_0 = 0 \quad \rightsquigarrow \quad \text{eq. ★ is trivially satisfied}$$

2) If “ A ” corresponds to a broken direction

$$T^A \equiv X^\alpha, \quad X^\alpha \phi_0 \neq 0 \quad \rightsquigarrow \quad \text{eq. ★ can be interpreted as follows}$$

$$\frac{\partial^2 V}{\partial \phi^a \partial \phi^c}(\phi_0) \underbrace{(X^\alpha \phi_0)^a}_{\neq 0} \stackrel{!}{=} 0$$

This equation is telling us that $X^\alpha \phi_0$ is an eigenvector of the mass matrix with zero eigenvalue (massless field).

Added explanation (not in the original notes)

This is the whole theorem, in one equation. Everything before it — the invariance of V , the differentiation, the split of the generators — was preparation. Equation ★ says: *for every generator, the vector $T^A \phi_0$ in field space is annihilated by the mass matrix.* For unbroken generators the vector is zero and the statement is empty. For broken ones it is a genuine non-zero vector, and a non-zero vector annihilated by M^2 is precisely a null eigenvector — a direction in field space costing no energy, hence a massless particle.

Where the two vanishing pieces come from. The second term of the first line contains $\partial V / \partial \phi^a(\phi_0)$, which is zero because ϕ_0 is a stationary point of V — the same tadpole condition used in Lecture 24 [p. 12*****]. The first term survives and is the mass matrix contracted with $T^A \phi_0$.

The geometric translation, which is the version worth remembering. $X^\alpha \phi_0$ is the infinitesimal displacement of the vacuum under a broken symmetry transformation — it is a vector *tangent to the vacuum manifold* at ϕ_0 . Since every point of that manifold has the same energy, V is exactly flat along it, so its Hessian must annihilate every tangent direction. Zero curvature = zero eigenvalue = massless mode. The algebra above is nothing but this sentence written in indices.

What is *not* being claimed. The theorem gives a lower bound: at least $\dim G - \dim H$ zero eigenvalues. A potential could accidentally have additional flat directions not related to any symmetry, producing extra massless scalars; those are not Goldstone bosons and are not protected — radiative corrections generically lift them. The symmetry-guaranteed ones cannot be lifted at any order.

2 The Goldstone theorem [p. 44*****]

Since we can repeat this reasoning for all “ α ” running over broken generators, we conclude:

There is **one Goldstone boson** (massless scalar) for every linearly independent spontaneously broken generator. **“Goldstone theorem”**

$$\# \begin{matrix} \text{Goldstone} \\ \text{bosons} \end{matrix} = \dim G - \dim H$$

Let us apply the theorem in two notable examples.

Added explanation (not in the original notes)

Why “linearly independent”. If some combination $c_\alpha X^\alpha$ annihilated ϕ_0 , that combination would by definition be an unbroken generator and would already belong to $\mathfrak{h}$. So the vectors $\{X^\alpha \phi_0\}$ are linearly independent, and there are exactly $\dim G - \dim H$ of them — hence at least that many zero eigenvalues, one per broken direction.

The map is not always injective in general, but it is here. Strictly the theorem counts the *independent vectors* $X^\alpha \phi_0$, and one should check that different broken generators give different tangent vectors. That is guaranteed by the definition of the split: the kernel of $T \mapsto T\phi_0$ is $\mathfrak{h}$, so the map descends to an isomorphism between $\mathfrak{g}/\mathfrak{h}$ and the tangent space to the vacuum manifold. This is the statement that [p. 52*****] is heading towards.

Historical note and scope. The result is due to Nambu (1960) and Goldstone (1961), with the general proof by Goldstone, Salam and Weinberg (1962). It requires: a *continuous* symmetry (a discrete one has no generators and gives no Goldstones — breaking $\mathbb{Z}_2$ produces domain walls, not massless particles); a *global* symmetry (gauging it changes everything — the Goldstones are eaten, which is the next lecture); Lorentz invariance (in non-relativistic systems the counting can differ, as for ferromagnets, where two broken generators give a single magnon); and a genuine ground state degeneracy in infinite volume.

PART II — Example: $SO(N) \rightarrow SO(N-1)$ [p. 44*****]–[p. 47*****]

3 Setup: N real scalars in the fundamental of $SO(N)$ [p. 44*****]

Example: $SO(N) \rightarrow SO(N-1)$

Consider a multiplet of N real scalar fields, transforming according to the fundamental representation of $SO(N)$.

$$\phi(x) = \begin{pmatrix} \phi_1(x) \\ \vdots \\ \phi_N(x) \end{pmatrix} \quad \begin{aligned} \phi(x) &\rightarrow R(\vec{\theta}) \phi(x) \\ \det R &= 1, \quad R^T R = R R^T = \mathbb{1} \end{aligned}$$

$$\dim G = \dim SO(N) = \frac{N(N-1)}{2}$$

We consider the invariant Lagrangian

Added explanation (not in the original notes)

Where $\dim SO(N) = N(N-1)/2$ comes from. An $SO(N)$ element is a real $N \times N$ matrix with $R^T R = \mathbb{1}$ and $\det R = 1$. The constraint $R^T R = \mathbb{1}$ is a symmetric matrix equation, so it imposes $N(N+1)/2$ conditions on the N^2 entries, leaving $N^2 - N(N+1)/2 = N(N-1)/2$. Equivalently: the generators are real antisymmetric $N \times N$ matrices, and there are $N(N-1)/2$ of those — one independent rotation per plane (i, j) with $i < j$. Sanity checks: $SO(2)$ gives 1 (Lecture 24), $SO(3)$ gives 3.

This is Lecture 24 with N instead of 2. The structure of the argument will be identical — invariant $\phi^T \phi$, Mexican-hat potential, degenerate minimum, choose a vacuum, shift, read off masses. The only new feature is that the vacuum manifold is now a sphere rather than a circle, and the unbroken subgroup is non-trivial.

4 The vacuum manifold S^{N-1} and the unbroken $SO(N-1)$ [p. 45*****]

$$\begin{aligned} \mathcal{L} &= \frac{1}{2} (\partial_\mu \phi)^T (\partial^\mu \phi) - V(\phi^T \phi) \\ V(\phi^T \phi) &= -\frac{\mu^2}{2} \phi^T \phi + \frac{\lambda}{4} (\phi^T \phi)^2 \end{aligned} \quad \phi^T \phi = \phi_1^2 + \cdots + \phi_N^2$$

$\hookrightarrow$ SSB is triggered by $\mu^2 > 0, \lambda > 0$

The minima of the potential are described by the condition

$$\phi_1^2 + \cdots + \phi_N^2 = \frac{\mu^2}{\lambda} = v^2$$

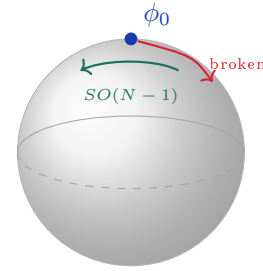
The vacuum is degenerate.

In field space, this is an equation for a S^{N-1} -sphere.

↪ The vacuum manifold of the theory is the $(N-1)$ -dimensional sphere.

We make a specific choice of the vacuum. For instance, we take

$$\langle \phi \rangle = \phi_0 = \begin{pmatrix} 0 \\ \vdots \\ \vdots \\ v \end{pmatrix}$$



the vacuum manifold S^{N-1} , drawn for $N=3$

This choice makes manifest the presence of an unbroken symmetry: the vacuum is left invariant if we rotate the first $(N-1)$ components.

The unbroken symmetry, therefore, is $SO(N-1)$.

Added explanation (not in the original notes)

Why the sphere. The minimum condition fixes only the *length* of the vector ϕ in field space, not its direction. The set of real N -vectors of fixed length v is by definition the sphere S^{N-1} of radius v , a manifold of dimension $N-1$. For $N=2$ this is the circle S^1 of Lecture 24; for $N=3$ it is the ordinary two-sphere drawn above.

Why the unbroken group is $SO(N-1)$. With ϕ_0 pointing along the N -th axis, a rotation leaves it fixed precisely when it acts only in the $(N-1)$ -dimensional subspace orthogonal to that axis — these are the rotations of the first $N-1$ components, forming $SO(N-1)$, of dimension $(N-1)(N-2)/2$. Every other generator moves ϕ_0 somewhere else on the sphere and is therefore broken. In the picture: the arrow along the latitude circle through the north pole shrinks to nothing there (unbroken), while displacements along the surface away from the pole are the broken directions.

The choice of pole is free. Any point of S^{N-1} would do, and choosing the north pole is the analogue of choosing $(v, 0)^T$ in Lecture 24 [p. 9*****]. Different choices give conjugate copies of $SO(N-1)$ — which is exactly the statement, from Lecture 25 [p. 41*****], that H is not a normal subgroup.

5 Counting the Goldstones; shifted fields [p. 46*****]

Goldstone theorem, therefore, implies the existence of $\dim G - \dim H = N - 1$ Goldstone bosons.

$$\begin{aligned} \dim G - \dim H &= \frac{N(N-1)}{2} - \frac{(N-1)(N-2)}{2} = \frac{N^2 - N - N^2 + 2N + N - 2}{2} \\ &= \frac{2(N-1)}{2} = N - 1. \end{aligned}$$

Let us check explicitly their existence.

We introduce the shifted fields

$$\phi(x) = \begin{pmatrix} \theta_1(x) \\ \vdots \\ \vdots \\ v + \chi(x) \end{pmatrix}$$

and write the Lagrangian in the broken phase. We write

$$\phi^T \phi = \underbrace{\sum_{a=1}^{N-1} \theta_a^2}_{\equiv \theta^2} + (v + \chi)^2 = \theta^2 + v^2 + \chi^2 + 2v\chi$$

We thus have at the Lagrangian level

$$\mathcal{L} = \frac{1}{2} \sum_{a=1}^{N-1} (\partial_\mu \theta_a)(\partial^\mu \theta_a) + \frac{1}{2} (\partial_\mu \chi)(\partial^\mu \chi) - V(\theta_a, \chi)$$

where the potential is:

Added explanation (not in the original notes)

The counting, done more transparently. Rather than expanding the polynomials, note that $\dim SO(N) - \dim SO(N-1)$ counts the generators of $SO(N)$ that are *not* rotations of the first $N-1$ axes among themselves — that is, the rotations in the planes (i, N) for $i = 1, \dots, N-1$. There are exactly $N-1$ of them, one for each direction one can tip the north pole. And $N-1 = \dim S^{N-1}$: the number of Goldstones equals the dimension of the vacuum manifold, which is the observation the notes will return to on [p. 52*****].

Degree-of-freedom bookkeeping. We started with N real fields. One combination — the radial one, χ — climbs the wall of the potential and is massive. The remaining $N-1$ — the θ_a , tangent to the sphere — are the Goldstones. Total preserved: $1 + (N-1) = N$. Shifting fields never creates or destroys degrees of freedom; it only reorganizes them into mass eigenstates.

Check against Lecture 24. $N = 2$: one Goldstone and one massive scalar, the vacuum manifold is S^1 , and H is trivial ($\dim SO(1) = 0$). Exactly the model of Lecture 24.

6 The potential in the broken phase; the $N-1$ Goldstones

[p. 47*****]

$$V(\theta_a, \chi) = -\frac{\mu^2}{2} (\theta^2 + v^2 + \chi^2 + 2v\chi) + \frac{\lambda}{4} (\theta^2 + v^2 + \chi^2 + 2v\chi)^2$$

⇓ This is the same potential that we analyzed in the case of $SO(2)$, with now $\theta^2 \equiv \sum_{a=1}^{N-1} \theta_a^2$ instead of being a single field.

The math is however identical, and we get

$$V(\theta_a, \chi) = V_0 + \frac{1}{2} (2\mu^2) \chi^2 + v\lambda \chi (\theta^2 + \chi^2) + \frac{\lambda}{4} (\theta^2 + \chi^2)^2$$

We see that θ_a with $a = 1, \dots, N-1$ are Goldstone fields (massless) of the breaking pattern

$$SO(N) \longrightarrow SO(N-1)$$

Added explanation (not in the original notes)

Deriving the second box. Write $S \equiv \theta^2 + \chi^2 + 2v\chi$, so that $\phi^T \phi = v^2 + S$. Then

$$V = -\frac{\mu^2}{2}(v^2 + S) + \frac{\lambda}{4}(v^2 + S)^2 = \underbrace{\left[-\frac{\mu^2 v^2}{2} + \frac{\lambda v^4}{4}\right]}_{V_0} + S \underbrace{\left[-\frac{\mu^2}{2} + \frac{\lambda v^2}{2}\right]}_{=0} + \frac{\lambda}{4}S^2,$$

the linear-in- S term vanishing because $\lambda v^2 = \mu^2$. Expanding $S^2 = (\theta^2 + \chi^2)^2 + 4v\chi(\theta^2 + \chi^2) + 4v^2\chi^2$ and using $\lambda v^2 = \mu^2$ on the last piece gives $\lambda v^2\chi^2 = \mu^2\chi^2 = \frac{1}{2}(2\mu^2)\chi^2$ — exactly the boxed result, and $V_0 = -\mu^4/4\lambda$.

Read off the spectrum. The quadratic part of V is $\frac{1}{2}(2\mu^2)\chi^2$ and *nothing else*: no θ_a^2 term appears anywhere, because every occurrence of θ comes in the combination θ^2 multiplied by at least one more field. So $m_\chi^2 = 2\mu^2$ — the same value as in Lecture 24 — and all $N - 1$ fields θ_a are exactly massless. The theorem is confirmed by direct computation.

A residual symmetry visible in the answer. The potential depends on the θ_a only through $\theta^2 = \sum_a \theta_a^2$, which is manifestly $SO(N - 1)$ invariant. That is the unbroken subgroup acting linearly on the Goldstone multiplet — the general statement from Lecture 25 [p. 41 *****] that $[T^a, X^\alpha] \sim X^\beta$, i.e. the broken generators (and hence the Goldstones) form a representation of H . Here the $N - 1$ Goldstones sit in the fundamental of $SO(N - 1)$.

PART III — Example: $SU(N) \rightarrow SU(N-1)$ [p. 47*****]–[p. 51*****]

7 Setup: N complex scalars in the fundamental of $SU(N)$ [p. 47*****]–[p. 48*****]

Example: $SU(N) \rightarrow SU(N-1)$

We now consider the case of N complex fields, transforming according to the fundamental representation of $SU(N)$.

$$\Phi(x) = \begin{pmatrix} \Phi_1(x) \\ \vdots \\ \Phi_N(x) \end{pmatrix} \qquad \begin{aligned} \Phi(x) &\longrightarrow U(\vec{\alpha}) \Phi(x) \\ \det U &= 1, \quad U^\dagger U = U U^\dagger = \mathbb{1} \end{aligned}$$

$$\dim G = \dim SU(N) = N^2 - 1$$

We consider the invariant Lagrangian

$$\begin{aligned} \mathcal{L} &= (\partial_\mu \Phi^\dagger)(\partial^\mu \Phi) - V(\Phi^\dagger \Phi) \\ V(\Phi^\dagger \Phi) &= -\mu^2 \Phi^\dagger \Phi + \lambda (\Phi^\dagger \Phi)^2 \end{aligned} \qquad \Phi^\dagger \Phi = \Phi_1^* \Phi_1 + \cdots + \Phi_N^* \Phi_N$$

$\hookrightarrow$ SSB is triggered if $\mu^2 > 0, \lambda > 0$

Added explanation (not in the original notes)

Where $\dim SU(N) = N^2 - 1$ comes from. A generic complex $N \times N$ matrix has $2N^2$ real parameters. Unitarity $U^\dagger U = \mathbb{1}$ is a hermitian matrix condition, imposing N^2 real constraints, leaving N^2 — that is $U(N)$. The further condition $\det U = 1$ removes one more, giving $N^2 - 1$. Equivalently the generators are traceless hermitian $N \times N$ matrices, of which there are $N^2 - 1$. Checks: $SU(2)$ gives 3, $SU(3)$ gives 8 (the Gell-Mann matrices).

The factor-of-two bookkeeping, once and for all. A complex field carries two real degrees of freedom, and this shows up in three places at once. (i) The kinetic term for a complex field has coefficient 1, not $\frac{1}{2}$, precisely so that each real component ends up canonically normalized. (ii) Correspondingly the potential is written $-\mu^2 \Phi^\dagger \Phi + \lambda (\Phi^\dagger \Phi)^2$, without the $\frac{1}{2}$ and $\frac{1}{4}$ of the real case — compare [p. 45*****]. (iii) As a result the minimum sits at $\Phi^\dagger \Phi = \mu^2/2\lambda$ rather than at $\phi^T \phi = \mu^2/\lambda$. All three conventions are tied together; none of them is a misprint, and this is exactly the same relation between the real and complex formulations that appeared in Lecture 24 [p. 15*****].

Why SU rather than U . $U(N) = SU(N) \times U(1)$ up to a discrete quotient, and the extra overall $U(1)$ phase would also be broken by the vacuum, contributing one more Goldstone. The notes restrict to $SU(N)$, where the $\det U = 1$ condition removes it. Keep this in mind: the counting below, $2N - 1$, would become $2N$ for $U(N)$.

8 The vacuum manifold S^{2N-1} and the unbroken $SU(N-1)$

[p. 48*****]–[p. 49*****]

The minima of the potential are described by the condition

$$|\Phi_1|^2 + \cdots + |\Phi_N|^2 = \frac{v^2}{2} = \frac{\mu^2}{2\lambda}$$

The vacuum is degenerate.

In field space, this is the equation for a $(2N-1)$ -dimensional sphere.

↪ The vacuum manifold of the theory is the $(2N-1)$ -dimensional sphere.

We make a specific choice for the vacuum. We take

$$\langle \Phi \rangle = \Phi_0 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ \vdots \\ v \end{pmatrix}$$

This choice makes manifest the presence of an unbroken symmetry: the vacuum is left invariant if we act on it with an $SU(N-1)$ transformation acting on the first $N-1$ components.

The unbroken symmetry, therefore, is $SU(N-1)$.

The Goldstone theorem, therefore, implies the existence of $\dim G - \dim H = 2N - 1$ Goldstone bosons.

$$\dim G - \dim H = N^2 - 1 - [(N-1)^2 - 1] = N^2 - 1 - (N^2 - 2N + 1) = 2N - 1.$$

Added explanation (not in the original notes)

Why a $(2N-1)$ -sphere. $\sum_k |\Phi_k|^2$ is the squared length of a vector in $\mathbb{C}^N \cong \mathbb{R}^{2N}$: writing $\Phi_k = x_k + iy_k$, the condition reads $\sum_k (x_k^2 + y_k^2) = v^2/2$, a sphere in $2N$ real dimensions, hence of dimension $2N-1$. For $N=1$ this is S^1 , the circle of Lecture 24 — as it must be, since $SU(1)$ is trivial and one recovers the abelian case only after restoring the overall phase.

A subtlety in the counting worth pausing on. $\dim S^{2N-1} = 2N-1$ and $\dim G - \dim H = 2N-1$: they agree. But note that $SU(N)/SU(N-1) \cong S^{2N-1}$ is a genuine group-theoretic fact (the fundamental of $SU(N)$ acts transitively on the unit sphere in $\mathbb{C}^N$), not a coincidence of dimensions. This equality of the two numbers is exactly what [p. 52*****] will highlight.

Why the stabilizer is $SU(N-1)$ and not $U(N-1)$. A transformation fixing $\Phi_0 \propto (0, \dots, 0, 1)^T$ must be block diagonal, acting arbitrarily on the first $N-1$ components and leaving the last one alone — so of the form $\text{diag}(W, 1)$ with $W \in U(N-1)$. But $\det U = 1$ forces $\det W = 1$, hence $W \in SU(N-1)$, of dimension $(N-1)^2 - 1$. (Had we started from $U(N)$, the stabilizer would have been the larger $U(N-1)$, and one would find $2N$ Goldstones — consistent with the extra broken $U(1)$.)

9 Shifted fields and the invariant $\Phi^\dagger \Phi$ [p. 49*****]–[p. 50*****]

Let's check explicitly their existence.

We introduce the shifted fields

$$\Phi(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} \rho_1(x) \\ \vdots \\ \rho_{N-1}(x) \\ v + \rho_N(x) \end{pmatrix} \quad \longrightarrow \text{Each } \rho_i(x) \text{ is a complex scalar field ("} v \text{" is a real parameter).}$$

We start by looking at the invariant quantity $\Phi^\dagger \Phi$. We write

$$\begin{aligned} 2 \Phi^\dagger \Phi &= (\rho_1^*, \dots, \rho_{N-1}^*, v + \rho_N^*) \begin{pmatrix} \rho_1 \\ \vdots \\ \rho_N + v \end{pmatrix} = \\ &= \sum_{k=1}^{N-1} |\rho_k|^2 + (v + \rho_N^*)(v + \rho_N) = \\ &= \sum_{k=1}^{N-1} |\rho_k|^2 + (v^2 + v\rho_N + v\rho_N^* + |\rho_N|^2) = \\ &= \sum_{k=1}^N |\rho_k|^2 + v^2 + v(\rho_N + \rho_N^*) \end{aligned}$$

$\rightsquigarrow$

$$2 \Phi^\dagger \Phi = \sum_{k=1}^N |\rho_k|^2 + v^2 + v(\rho_N + \rho_N^*)$$

Added explanation (not in the original notes)

Slip flagged. The handwritten line labels this computation $\Phi^\dagger \Phi$, but the row and column vectors written out are $\sqrt{2} \Phi^\dagger$ and $\sqrt{2} \Phi$ — the explicit $1/\sqrt{2}$ in the definition of Φ has been dropped. What is being computed is therefore $2 \Phi^\dagger \Phi$, which is precisely how the boxed result is labelled one line later. The notes self-correct; I have written the factor of 2 in from the first line so that the chain reads consistently.

Why $1/\sqrt{2}$ in the definition at all. It makes each real component canonically normalized. With $\Phi = \frac{1}{\sqrt{2}}(\rho_1, \dots)^T$ the kinetic term $(\partial \Phi^\dagger)(\partial \Phi) = \frac{1}{2} \sum_k |\partial \rho_k|^2$, and writing $\rho_k = a_k + ib_k$ gives $\frac{1}{2} \sum_k [(a_k)^2 + (b_k)^2]$ — the standard normalization for $2N$ real scalars. The same convention was used in Lecture 24 [p. 15*****] for the single complex field.

Note the structure of the answer. It splits into a piece quadratic in all the fields ($\sum_k |\rho_k|^2$), a constant (v^2), and — the important one — a piece *linear* in a single real combination, $v(\rho_N + \rho_N^*) = 2v \operatorname{Re} \rho_N$. Only that one real direction is singled out by the vacuum, and it is the one that will end up massive. Every other real component appears only quadratically, and will drop out of the quadratic potential once the tadpole cancellation is used.

10 The quadratic potential: only one field gets a mass [p. 50*****]— [p. 51*****]

To simplify the algebra, since we are interested in the mass spectrum, we will only keep in the following terms up to order quadratic in the fields.

$$\begin{aligned}
(2\Phi^\dagger\Phi)^2 &= \left[\sum_{k=1}^N |\rho_k|^2 + v^2 + v(\rho_N + \rho_N^*) \right]^2 = \\
&= \underbrace{2v^2 \sum_{k=1}^N |\rho_k|^2 + v^2(\rho_N + \rho_N^*)^2 + \dots}_{\text{terms quadratic in the fields}}
\end{aligned}$$

The potential reads (only quadratic terms)

$$\begin{aligned}
V &= -\frac{\mu^2}{2} \left(\sum_{k=1}^N |\rho_k|^2 \right) + \frac{\lambda}{4} \left[2v^2 \sum_{k=1}^N |\rho_k|^2 + v^2(\rho_N + \rho_N^*)^2 \right] + \dots \\
&= \left(\sum_{k=1}^N |\rho_k|^2 \right) \underbrace{\left(-\frac{\mu^2}{2} + \frac{\lambda v^2}{2} \right)}_{-\frac{\mu^2}{2} + \frac{\lambda}{2} \left(\frac{\mu^2}{\lambda} \right) = 0} + \underbrace{\frac{\lambda v^2}{4} (\rho_N + \rho_N^*)^2}_{\frac{\lambda}{4} \left(\frac{\mu^2}{\lambda} \right) (\rho_N + \rho_N^*)^2} + \dots
\end{aligned}$$

We thus get

$$V = \frac{\mu^2}{4} (\rho_N + \rho_N^*)^2 + \dots$$

The outcome is that only one real scalar field takes a mass, $(\rho_N + \rho_N^*) \equiv 2 \operatorname{Re}(\rho_N)$.

We started with $2N$ (real) d.o.f., and only 1 takes a mass $\rightsquigarrow (2N - 1)$ d.o.f. remain massless, and these are the Goldstone bosons.

Added explanation (not in the original notes)

Reading off the mass. Write $\rho_N = \frac{1}{\sqrt{2}}(h + i\pi_N)$ with h, π_N real and canonically normalized. Then $\rho_N + \rho_N^* = \sqrt{2}h$ and

$$V = \frac{\mu^2}{4} \cdot 2h^2 + \dots = \frac{1}{2}(2\mu^2)h^2 + \dots,$$

so $m_h^2 = 2\mu^2$ — the same universal answer as in Lecture 24 and in the $SO(N)$ example. That is no coincidence: in every one of these models the massive mode is the radial fluctuation, and its mass² is V'' at the minimum of the same Mexican-hat profile.

Why v^2 and not $v^2/2$ in the quadratic terms. From $\sum_k |\Phi_k|^2 = v^2/2$ one has $2\Phi^\dagger\Phi|_{\text{vac}} = v^2$, which is why squaring the bracket produces the cross term $2 \cdot v^2 \cdot \sum_k |\rho_k|^2$. And $\lambda v^2 = \mu^2$ follows from $\mu^2/2\lambda = v^2/2$ — consistent with the real case despite the different-looking conventions.

The degree-of-freedom ledger. N complex fields = $2N$ real d.o.f. One real combination, $\operatorname{Re}(\rho_N)$, is massive; the other $2N - 1$ — namely $\operatorname{Im}(\rho_N)$ and the $2(N - 1)$ real components of $\rho_1, \dots, \rho_{N-1}$ — are massless. That matches $\dim G - \dim H = 2N - 1$ exactly, confirming the theorem in a second, quite different family of models.

Sanity check at $N = 1$. Then there are 2 real d.o.f., one massive and one massless — precisely the h and π of Lecture 24. (As noted above, $SU(1)$ is trivial, so this case is really the $U(1)$ model; the arithmetic still works because the broken generator is the overall phase.)

PART IV — Comments, and Goldstone bosons & geometry

[p. 51*****]–[p. 52*****]

11 Comments and observations [p. 51*****]–[p. 52*****]

Comments and observations

1) The breaking pattern $SU(N) \rightarrow SU(N-1)$ depends crucially on the fact that Φ transforms according to the fundamental.

2) In both the examples we have discussed, the Goldstone bosons still appear in the potential (beyond the quadratic order).

↔ This is because our choice of field variables is not ideal to make transparent the properties of Goldstone bosons. It would be nice to find something equivalent to the exponential parametrization used in the case of $SO(2)$.

3) Is there a connection between the number of Goldstone bosons and the dimension of the vacuum manifold?

↔ In the two examples we have discussed,

$$\# \text{ Goldstone bosons} = \text{dimension of the vacuum manifold}$$

Added explanation (not in the original notes)

On comment 1. The pattern of breaking is not a property of the group alone — it depends on which representation the scalars live in, because that determines which generators can annihilate a given ϕ_0 . For $SU(N)$ with Φ in the fundamental, a vacuum is a vector in $\mathbb{C}^N$ and its stabilizer is $SU(N-1)$. Put the scalars in the *adjoint* instead (an $N \times N$ traceless hermitian matrix) and a typical vacuum can be diagonalized, leaving $SU(N) \rightarrow SU(N-1) \times U(1)$, or with more degenerate eigenvalues $SU(N) \rightarrow SU(p) \times SU(q) \times U(1)$ with $p+q=N$. Same group, different representation, completely different pattern — and a different number of Goldstones. This is why model building consists largely of choosing representations.

On comment 2 — and where it leads. In Lecture 24 the exponential parametrization $\varrho = \frac{1}{\sqrt{2}}(v+h)e^{i\pi/v}$ removed π from the potential entirely, making its shift symmetry and derivative couplings manifest. The Cartesian variables used in this lecture do not: the θ_a still appear in V at cubic and quartic order (see the boxed potential on [p. 47*****]). Nothing is *wrong* — the masses are right and the symmetry is intact — but the structure is hidden. The generalization being hinted at is

$$\Phi(x) = e^{i\pi^\alpha(x)X^\alpha/f} (\Phi_0 + \text{radial modes}),$$

i.e. exponentiate the *broken* generators. This is the CCWZ construction (Callan, Coleman, Wess, Zumino 1969) mentioned in Lecture 24 [p. 21*****]; it makes the Goldstones co-ordinates on G/H by construction, and guarantees that they appear only with derivatives.

On comment 3 — the answer is yes, and it is not an accident. $SO(N) \rightarrow SO(N-1)$: the vacuum manifold is S^{N-1} , of dimension $N-1$, and there are $N-1$ Goldstones. $SU(N) \rightarrow SU(N-1)$: the manifold is S^{2N-1} , of dimension $2N-1$, and there are $2N-1$ Goldstones. The next section explains why this must always hold.

12 Goldstone bosons and geometry [p. 52*****]

■ Goldstone bosons and geometry (point 3)

Consider partial SSB of the type $G \rightarrow H$. We have a set of scalar fields Φ that experience some potential $V(\Phi)$ that is minimized by the vacuum manifold.

$$\mathcal{M}_0 \equiv \left\{ \Phi : V(\Phi) = V_{\min} \right\}$$

vacuum manifold

✓ Different field configurations that belong to $\mathcal{M}_0$ are related by G -transformations.

↘ If we select a specific element of $\mathcal{M}_0$, say Φ_0 , the H -transformations leave Φ_0 invariant.

[the scan ends here]

Added explanation (not in the original notes) — Completing the argument — the vacuum manifold is the coset G/H

The two green annotations are precisely the two hypotheses needed; here is the conclusion they were leading to.

Step 1 — the action is transitive. The first annotation says that any two points of $\mathcal{M}_0$ are related by some $g \in G$. In the language of group actions, $\mathcal{M}_0$ is a single *orbit* of G . (This is a genuine assumption about the potential, and it holds in both examples of this lecture: $SO(N)$ acts transitively on S^{N-1} , and $SU(N)$ on S^{2N-1} .)

Step 2 — the stabilizer is H . The second annotation says that the subgroup fixing a chosen Φ_0 is exactly H ; by definition, it is generated by the unbroken generators.

Step 3 — put them together. Consider the map $g \mapsto g \Phi_0$ from G to $\mathcal{M}_0$. By Step 1 it is surjective. Two elements g_1, g_2 give the same point iff $g_2^{-1} g_1 \in H$, i.e. iff they lie in the same left coset gH . Therefore

$$\mathcal{M}_0 \cong G/H$$

as a manifold, and consequently

$$\dim \mathcal{M}_0 = \dim G - \dim H = \# \text{ Goldstone bosons.}$$

That is the answer to comment 3, and it is now a theorem rather than an observation drawn from two examples.

The tangent-space picture, which ties the whole chapter together. Near Φ_0 , a point of $\mathcal{M}_0$ is reached by acting with a broken generator: $\Phi_0 \rightarrow \Phi_0 - i\epsilon_\alpha X^\alpha \Phi_0$. The vectors $X^\alpha \Phi_0$ therefore *span the tangent space* to $\mathcal{M}_0$ at Φ_0 , and there are $\dim G - \dim H$ of them. But those are exactly the vectors that equation ★ on [p. 43*****] showed to be null eigenvectors of the mass matrix. So:

$$\text{tangent directions to } \mathcal{M}_0 \iff \text{flat directions of } V \iff \text{massless modes.}$$

The Goldstone fields are, quite literally, coordinates on the vacuum manifold. Everything in Lectures 24–26 — the circle of Lecture 24, the sphere of [p. 45*****], the algebra of [p. 43*****] — is one statement seen from different angles.

Checking the two examples. $SO(N)/SO(N-1) \cong S^{N-1}$ and $SU(N)/SU(N-1) \cong S^{2N-1}$: both are standard results, and both are visible directly — the group acts on the unit sphere in $\mathbb{R}^N$ (resp. $\mathbb{C}^N$) transitively, with the stabilizer of a point being the subgroup acting on the orthogonal complement.

Why this matters beyond bookkeeping. Once one knows the Goldstones live on G/H , the low-energy effective Lagrangian writes itself: it is the most general theory of maps from spacetime into G/H , whose leading term is the “non-linear sigma model” $\frac{f^2}{2} g_{\alpha\beta}(\pi) \partial_\mu \pi^\alpha \partial^\mu \pi^\beta$

with $g_{\alpha\beta}$ the G -invariant metric on the coset. The exponential parametrization asked for in comment 2 is exactly the choice of coordinates that makes this manifest, and the EFT constructed in Lecture 25 for a single Goldstone is its simplest instance.

Summary of Lecture 26 — the logic in eight lines

Added explanation (not in the original notes) — Summary

1. Evaluating the differentiated invariance condition at $\phi = \phi_0$ kills the tadpole term and leaves $M_{ac}^2 (T^A \phi_0)^a = 0$ for every generator — the equation ★. [p. 43*****]
2. For unbroken generators $T^a \phi_0 = 0$ and it is empty. For broken ones $X^a \phi_0 \neq 0$ is a non-zero null eigenvector of the mass matrix: a massless field. [p. 43*****]
3. Hence the **Goldstone theorem**: one massless scalar per linearly independent broken generator, $\# \text{Goldstones} = \dim G - \dim H$. [p. 44*****]
4. *Example 1*, $SO(N)$ with N real scalars in the fundamental, $\dim G = N(N-1)/2$. The minimum $\phi^T \phi = v^2$ is the sphere S^{N-1} ; choosing $\phi_0 = (0, \dots, 0, v)^T$ leaves $SO(N-1)$ unbroken, so there are $N-1$ Goldstones. [p. 44*****]–[p. 46*****]
5. Explicit check: shifting $\phi = (\theta_1, \dots, \theta_{N-1}, v + \chi)^T$ gives $V = V_0 + \frac{1}{2}(2\mu^2)\chi^2 + v\lambda\chi(\theta^2 + \chi^2) + \frac{\lambda}{4}(\theta^2 + \chi^2)^2$: one massive mode, $N-1$ exactly massless θ_a . [p. 46*****]–[p. 47*****]
6. *Example 2*, $SU(N)$ with N complex scalars in the fundamental, $\dim G = N^2 - 1$. The minimum is the sphere S^{2N-1} ; the stabilizer of $\Phi_0 \propto (0, \dots, 0, v)^T$ is $SU(N-1)$, so there are $2N-1$ Goldstones. [p. 47*****]–[p. 49*****]
7. Explicit check: keeping only quadratic terms, the tadpole coefficient $-\frac{\mu^2}{2} + \frac{\lambda v^2}{2}$ vanishes and $V = \frac{\mu^2}{4}(\rho_N + \rho_N^*)^2 + \dots$. Of $2N$ real d.o.f. exactly one, $\text{Re}(\rho_N)$, is massive with $m^2 = 2\mu^2$; the other $2N-1$ are the Goldstones. [p. 49*****]–[p. 51*****]
8. Three comments: the pattern depends on the *representation*, not just the group; the Cartesian variables hide the Goldstone structure and something like the exponential parametrization is wanted; and in both examples $\# \text{Goldstones} = \dim \mathcal{M}_0$. The last is explained by $\mathcal{M}_0 \cong G/H$. [p. 51*****]–[p. 52*****]

Added explanation (not in the original notes) — Where this is going

Two threads are left dangling, and both are picked up next.

The geometric one. $\mathcal{M}_0 \cong G/H$ means the Goldstone fields are coordinates on a coset space. Making that manifest requires exponentiating the broken generators, $\Phi = e^{i\pi^\alpha X^\alpha/f} \Phi_0 + \dots$ — the generalization of Lecture 24's $e^{i\pi/v}$ that comment 2 asks for. The payoff is the same as in the abelian case: the Goldstones acquire an exact shift symmetry, appear only through derivatives, and their low-energy Lagrangian is fixed by geometry alone.

The physical one. Everything in Lectures 24–26 concerned a *global* symmetry, and the Goldstones came out as genuine massless particles in the spectrum. Nature does not obviously contain such particles, and Lecture 23 left a gauge boson badly in need of a mass. Gauging the broken symmetry solves both problems at once: the would-be Goldstones become the longitudinal polarizations of the gauge bosons, one per broken generator, and the count $\dim G - \dim H$ becomes the number of massive vectors. In the Standard Model $SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{em}}$ gives $4 - 1 = 3$: the $W^\pm$ and the Z . That is the Higgs mechanism, and every ingredient it needs has now been assembled.